The Recognition Length λ_{rec}: A Conditional Derivation with Zero Adjustable Parameters

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Abstract

Within a discrete-lattice framework built on a uniquely determined cost functional $J(x) = \frac{1}{2}(x + x^{-1}) - 1$ and the combinatorics of the 3-cube Q_3, we derive a fundamental length scale $\lambda_{\text{rec}} = \sqrt{\pi}\,\ell_P \approx 2.86 \times 10^{-35}$ m from which Newton's gravitational constant follows: $G = c^3\lambda_{\text{rec}}^2/(\pi\hbar)$, with Einstein coupling $\kappa = 8\varphi^5$. The derivation has *zero continuously adjustable parameters*, but rests on three structural assumptions stated explicitly in the text: (i) a minimal-overhead balance condition equating informational and geometric costs, (ii) an identification of the Planck area with the inverse curvature load per mode via Gauss-Bonnet, and (iii) a decomposition of the curvature space into $|\kappa| = 4$ modes from Q_3 symmetry. Given these assumptions, λ_{rec}, G, and κ are fixed uniquely. The cost functional J is selected by a multiplicative d'Alembert composition law [1, 3, 2]. An independent lower bound via Bekenstein entropy saturation and Landauer erasure cost gives $\lambda_{\text{BL}} \approx 7.6 \times 10^{-36}$ m; the curvature-balanced scale exceeds this floor by a factor of $\sim$3.8, consistent with the lattice operating above the information-theoretic minimum. Two phenomenological cross-checks (stellar luminosity balance and vacuum energy density) agree at the 7% level. All prerequisite theorems and the balance-condition derivation are machine-verified in Lean 4 [10] with zero `sorry` statements.

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1 Introduction

1.1 The Problem: Deriving a Fundamental Length

The Standard Model of particle physics and General Relativity together contain approximately 26 free parameters—coupling constants, masses, mixing angles—that must be measured in the laboratory and inserted by hand. Among these, Newton's constant G sets the strength of gravity and, together with $\hbar$ and c, defines the Planck length $\ell_P = \sqrt{\hbar G/c^3} \approx 1.6 \times 10^{-35}$ m. Whether G is an irreducible empirical fact or derivable from something deeper remains open.

This paper shows that, given three explicit structural assumptions, a specific length scale $\lambda_{\text{rec}} = \sqrt{\pi}\,\ell_P$ is *uniquely fixed* with zero continuously adjustable parameters, and Newton's constant follows as a consequence: $G = c^3\lambda_{\text{rec}}^2/(\pi\hbar)$. The three ingredients are:

(i) A *cost functional* $J(x) = \frac{1}{2}(x + x^{-1}) - 1$, uniquely determined by a multiplicative d'Alembert composition law and calibration conditions (Theorem 2.3).

(ii) A *curvature cost* $J_{\text{curv}}(\lambda) = 2\lambda^2$, derived from the Gauss-Bonnet theorem applied to the bounding sphere of a causal diamond in $D = 3$, with curvature quanta inherited from the combinatorial structure of the 3-cube Q_3 (Proposition 4.1).

(iii) A *minimal-overhead balance condition* requiring the informational cost and the geometric curvature cost to be equal (Axiom; see Section 10.3).

Ingredient (i) is a theorem. Ingredient (ii) follows from Gauss-Bonnet plus a mode-counting argument. Ingredient (iii) is a physical postulate. The claim is not that G is derived from pure logic, but that within this framework there is no dial to turn: every constant is locked once the postulates are accepted.

1.2 The Scale-Invariance Problem

The cost functional $J(x)$ depends only on ratios: $J(\alpha a/\alpha b) = J(a/b)$ for any $\alpha > 0$. This scale invariance is why frameworks built on J can derive dimensionless constants (α^{-1}, mass ratios) but cannot, by the variational principle alone, select an absolute length. To fix a dimensional scale, one needs an argument that is not scale-invariant.

The curvature cost $J_{\text{curv}} = 2\lambda^2$ breaks the symmetry: it depends on λ^2, not on a ratio. Balanced against the fixed bit cost, it pins down a unique scale.

1.3 Context: Recognition Science

The derivation presented here arose within Recognition Science (RS), an information-theoretic framework that derives physical constants from two postulates: dual recognition (a definite physical state arises only when two distinct entities interact) and minimal informational overhead [5]. The cost functional $J(x)$ is uniquely selected by a d'Alembert composition law [1], with broader inevitability results establishing the same functional from general symmetric polynomial combiners [2] and axiomatic characterizations under inversion symmetry [3]. Observable structure is derived through a measurement-first quotient construction [4].

The RS forcing chain derives a sequence of constants from the golden ratio φ and the $D = 3$ cube topology:

$$\text{Postulates} \to J \to \varphi \to E_{\text{coh}} \to \tau_0 \to c \to \hbar \to \lambda_{\text{rec}} \to G \to \alpha^{-1} \to \text{mass spectrum}. \quad (1)$$

Every link before λ_{rec} produces a dimensionless output. The derivation of λ_{rec} bridges the dimensionless and dimensional sectors.

The mathematical content of this paper is self-contained and does not require familiarity with RS.

1.4 Historical Note

The dimensionless sector was established first [6]. Two empirical routes—stellar luminosity balance and the observed dark-energy density—provided independent estimates of an effective length scale, $\lambda_{\text{eff}} \approx 60\,\mu\text{m}$. These were cross-checks. A one-bit Bekenstein-Landauer argument provided the first internal derivation of the dimensional scale [7]. The curvature-extremum derivation presented here replaced that argument with one using only the cost functional and Gauss-Bonnet geometry. The Bekenstein-Landauer route is retained in Section 8 as an independent consistency floor.

1.5 Machine Verification

The prerequisite theorems and the balance-condition derivation have been formalized in Lean 4 using the Mathlib library. The complete codebase—122 Lean files, 32 verification certificates, zero `sorry` statements—is publicly available at

`https://github.com/jonwashburn/recognition-science`

and archived on Zenodo [10]. Module-level correspondence is recorded in Appendix A.

1.6 Outline

Section 2 states the mathematical prerequisites (cost functional, golden ratio, dimension forcing) with full proofs. Section 3 formulates the dimensional closure problem. Section 4 derives the curvature cost from the $D = 3$ cube structure and the Gauss-Bonnet theorem. Section 5 presents the main derivation. Section 6 carries out the dimensional restoration and derives Newton's constant. Section 7 discusses consequences for the forcing chain. Section 8 presents the independent Bekenstein-Landauer derivation as a cross-check. Section 9 gives phenomenological consistency tests. Section 10 discusses the result. Appendix A documents the Lean 4 verification.

2 Mathematical Prerequisites

Every theorem below is stated with a full proof outline. Lean 4 formalizations are recorded in Appendix A.

2.1 The Cost Functional

Definition 2.1 (Recognition Composition Law)**.** A function $F : \mathbb{R}_{>0} \to \mathbb{R}$ satisfies the *Recognition Composition Law* (RCL) if for all $x, y > 0$,

$$F(xy) + F(x/y) = 2\,F(x)\,F(y) + 2\,F(x) + 2\,F(y). \tag{2}$$

Definition 2.2 (Calibration conditions)**.** A function F is *normalized* if $F(1) = 0$, and *calibrated* if the log-coordinate reparametrization $G(t) := F(e^t)$ satisfies $G''(0) = 1$.

Theorem 2.3 (Uniqueness of J; Washburn-Zlatanović [1])**.** *Let $F : \mathbb{R}_{>0} \to \mathbb{R}$ satisfy:*

(i) *The RCL* (2)*,*

(ii) *Normalization: $F(1) = 0$,*

(iii) *Calibration: $G''(0) = 1$,*

(iv) *Continuity on $(0, \infty)$.*

Then F is uniquely determined:

$$F(x) = J(x) := \tfrac{1}{2}(x + x^{-1}) - 1 \qquad \textit{for all } x > 0. \tag{3}$$

Proof outline. Define $G(t) = F(e^t)$ and $H(t) = G(t) + 1$. Under the substitution $x = e^s$, $y = e^t$, the RCL (2) becomes the d'Alembert functional equation

$$H(s + t) + H(s - t) = 2\,H(s)\,H(t) \tag{4}$$

with $H(0) = 1$. Normalization gives $G(0) = 0$, hence $H(0) = 1$. Reciprocal symmetry $F(x) = F(x^{-1})$ (which follows from the RCL by setting $y = x$) gives $G(-t) = G(t)$, so H is even.

By Aczél's theorem on the d'Alembert equation, every continuous solution of (4) with $H(0) = 1$ that is not identically 1 is C^∞. This was proved by showing that continuous d'Alembert solutions satisfy the ODE $H'' = \alpha^2 H$ for some $\alpha \geq 0$, from which analyticity follows. (The Lean formalization uses the Aczél axiom to establish C^∞ regularity, then bootstraps to the ODE.)

The calibration $G''(0) = 1$ implies $H''(0) = 1$, so $\alpha^2 = 1$. Since H is even, $H'(0) = 0$. The unique solution to $H'' = H$, $H(0) = 1$, $H'(0) = 0$ is $H(t) = \cosh t$. This is proved by showing that any solution f with $f'' = f$, $f(0) = 0$, $f'(0) = 0$ must be identically zero (via the Gronwall-type argument: define $g = f' - f$ and $h = f' + f$, show $g' = -g$ and $h' = h$, use the integrating-factor method $\frac{d}{dt}[g\,e^t] = 0$ and $\frac{d}{dt}[h\,e^{-t}] = 0$ to conclude $g = h = 0$).

Therefore $G(t) = \cosh t - 1$, and $F(x) = G(\ln x) = \cosh(\ln x) - 1 = \frac{1}{2}(x + x^{-1}) - 1$. □

Remark 2.4 (Key properties of J). The cost functional satisfies:

- $J(x) = J(x^{-1})$ (reciprocal symmetry).
- $J(x) \geq 0$ for all $x > 0$, with equality iff $x = 1$. Proof: $J(x) = (x - 1)^2/(2x) \geq 0$.

- J is strictly monotone increasing on $[1, \infty)$.
- J is strictly convex on $\mathbb{R}_{>0}$.
- $J(1+\varepsilon) = \varepsilon^2/2 + O(\varepsilon^3)$ (quadratic approximation near equilibrium).

Remark 2.5 (Broader inevitability frame). Theorem 2.3 starts from the already reduced RCL form. A broader companion result, the d'Alembert inevitability theorem [2], begins with a general symmetric polynomial combiner

$$F(xy) + F(x/y) = P(F(x), F(y)), \qquad P \in \mathbb{R}[u, v], \quad \deg P \leq 2,$$

and proves that continuity and nonconstancy force

$$P(u, v) = 2u + 2v + c\,uv.$$

Under convexity, nonnegativity, and unit log-curvature calibration, the admissible solutions lie in the hyperbolic family

$$F_\alpha(x) = \frac{1}{\alpha^2}\big(\cosh(\alpha \ln x) - 1\big), \qquad \alpha \geq 1,$$

with $c = 2\alpha^2$. This provides a wider inevitability context for the canonical reciprocal cost used throughout the present paper.

2.2 Golden-Ratio Forcing

Definition 2.6 (Self-similar scale sequence)**.** A *geometric scale sequence* is a sequence $\{s_n\}$ with $s_{n+1} = r \cdot s_n$ for a fixed ratio $r > 0$, $r \neq 1$. It is *closed* if the compositional constraint of the discrete dynamics forces the recurrence $r^2 = r + 1$.

Theorem 2.7 (Phi forcing)**.** *Let $r > 0$ satisfy $r^2 = r + 1$. Then $r = \varphi = (1 + \sqrt{5})/2$.*

Proof. The equation $r^2 - r - 1 = 0$ has roots $(1 \pm \sqrt{5})/2$. Since $(1 - \sqrt{5})/2 < 0$ and $r > 0$, the unique positive solution is φ. The Lean proof verifies this by showing that for any $r > 0$ satisfying the constraint, $r = \varphi$ follows from the identity $(r - \varphi)^2 + (r + \varphi - 1)^2 = 0$ (which uses $\varphi^2 = \varphi + 1$ and the hypothesis $r^2 = r + 1$, forcing $r = \varphi$). □

Lemma 2.8 (Fibonacci power identities)**.** *The golden ratio satisfies $\varphi^n = F_n\,\varphi + F_{n-1}$ for all $n \geq 1$, where F_n is the n-th Fibonacci number. In particular:*

$$\begin{aligned} \varphi^2 &= \varphi + 1, & \varphi^3 &= 2\varphi + 1, \\ \varphi^4 &= 3\varphi + 2, & \varphi^5 &= 5\varphi + 3 \approx 11.09. \end{aligned} \tag{5}$$

Proof. By induction, using $\varphi^{n+1} = \varphi \cdot \varphi^n = \varphi(F_n\,\varphi + F_{n-1}) = F_n\,\varphi^2 + F_{n-1}\,\varphi = F_n(\varphi + 1) + F_{n-1}\,\varphi = (F_n + F_{n-1})\varphi + F_n = F_{n+1}\,\varphi + F_n$. Each case ($n = 2$ through $n = 5$) is verified individually in Lean by substituting $\varphi^2 = \varphi + 1$ and simplifying. □

Corollary 2.9 (Bounds)**.** $1.618 < \varphi < 1.619$ *and* $10.7 < \varphi^5 < 11.3$.

2.3 Eight-Tick Cycle and $D = 3$

Definition 2.10 (Eight-tick structure)**.** The 8-tick clock has phases $\theta_k = k\pi/4$ for $k = 0, 1, \ldots, 7$, with complex amplitudes $\exp(i\theta_k)$. The period is 8: $[\exp(i\theta_k)]^8 = \exp(2\pi i k) = 1$.

Theorem 2.11 ($D = 3$ forcing)**.** *Let D be a positive integer such that the minimal compatible walk period on the hypercube $Q_D = \{0,1\}^D$ is 2^D, and this period equals 8. Then $D = 3$.*

Proof. $2^D = 8$ implies $D = 3$. The Lean proof verifies this by case analysis: $D = 0, 1, 2$ give $2^D = 1, 2, 4 \neq 8$; $D = 3$ gives $2^3 = 8$; $D \geq 4$ gives $2^D \geq 16 > 8$.

Three independent arguments confirm that $D = 3$ is the unique compatible dimension:

1. **Eight-tick:** $2^D = 8 \Longrightarrow D = 3$.
2. **Spinor structure:** The Clifford algebra $\mathrm{Cl}_3 \cong M_2(\mathbb{C})$ gives 2-component complex spinors; $\mathrm{Spin}(3) \cong \mathrm{SU}(2)$ is the simplest non-abelian compact Lie group. No other D has all three properties (2-component, non-abelian, simple).
3. **Gap-45 synchronization:** $\mathrm{lcm}(8, 45) = 360 = 2^3 \times 3^2 \times 5$, and 2^3 identifies $D = 3$.

□

2.4 Prior Derived Constants

With $\ell_0 = 1$ and $\tau_0 = 1$ as the natural base units (one spatial quantum and one temporal quantum), the derivation sequence establishes:

Definition 2.12 (Natural units)**.**

$$c := \frac{\ell_0}{\tau_0} = 1 \qquad \text{(speed of light: one spatial quantum per time quantum)}, \tag{6}$$

$$E_{\mathrm{coh}} := \varphi^{-5} \qquad \text{(coherence energy from } \varphi\text{-scaling)}, \tag{7}$$

$$\hbar := E_{\mathrm{coh}} \cdot \tau_0 = \varphi^{-5} \qquad \text{(IR gate identity)}, \tag{8}$$

$$J_{\mathrm{bit}} := \ln \varphi \qquad \text{(information cost of one interaction event)}. \tag{9}$$

Lemma 2.13 (Properties of $\hbar$ in natural units)**.**

1. $\hbar > 0$ *(positivity: $\varphi > 0 \Longrightarrow \varphi^{-5} > 0$).*
2. $\hbar < 1$ *(sub-natural: $\varphi > 1 \Longrightarrow \varphi^5 > 1 \Longrightarrow \varphi^{-5} < 1$).*
3. $0.088 < \hbar < 0.093$ *(from $1.61 < \varphi < 1.62$ and the Fibonacci identity $\varphi^5 = 5\varphi + 3$).*

Proof. All three follow from the bounds on φ. For (3): $\varphi > 1.61$ gives $\varphi^5 > 1.61^5 \approx 10.82$, so $\hbar < 1/10.82 < 0.093$. Similarly, $\varphi < 1.62$ gives $\varphi^5 < 1.62^5 \approx 11.16$, so $\hbar > 1/11.16 > 0.088$. □

3 The Problem of Dimensional Closure

Definition 3.1 (Scale invariance of J)**.** The cost functional (3) depends only on the ratio x. For any $\alpha > 0$,

$$J\Big(\frac{\alpha\, a}{\alpha\, b}\Big) = J\Big(\frac{a}{b}\Big)\,. \tag{10}$$

The variational minimum at φ is therefore independent of any absolute scale.

Scale invariance is why the program derives dimensionless constants without specifying a length—and also why the variational principle alone cannot select a physical unit. The Planck length $\ell_P = \sqrt{\hbar G/c^3}$ involves G, which appears nowhere in the dimensionless sector.

The problem: *what internal argument selects a unique length scale?*

4 The Curvature Cost

4.1 The RS Token Alphabet on Q_3

The discrete lattice is a graph whose vertices are updated by interaction events. In $D = 3$, the local geometry is that of the 3-cube $Q_3 = \{0, 1\}^3$, which has 8 vertices, 12 edges, 6 faces (3 opposite face-pairs), and 1 solid cell.

Proposition 4.1 (Token curvature quanta)**.** *Each interaction event produces a token carrying $|\kappa| = 4$ independent curvature quanta.*

Proof. The symmetry group of Q_3 acts on the space of local curvature deformations. This space decomposes into four irreducible modes:

(i) Three *face-pair modes*, one for each of the three opposite face-pairs of Q_3 (the xy-, xz-, and yz-planes). Each mode carries curvature ± 1 representing compression or expansion along the corresponding normal direction.

(ii) One *scalar mode* associated with the Euler characteristic of Q_3. The combinatorial Gauss-Bonnet theorem gives each vertex of Q_3 an angle defect $\delta_v = 2\pi - 3(\pi/2) = \pi/2$, and the total defect $\sum_v \delta_v = 8 \times \pi/2 = 4\pi = 2\pi\chi(S^2)$ distributes evenly over the 4 modes.

The total unsigned curvature is $|\kappa| = |\pm 1| \times 4 = 4$. □

4.2 From Curvature to Cost via Gauss-Bonnet

An interaction event at spatial scale λ creates a causal diamond whose bounding surface is topologically S^2 with area $A = 4\pi\lambda^2$. The Gauss-Bonnet theorem for a closed orientable 2-surface Σ states

$$\int_\Sigma K\, dA = 2\pi\, \chi(\Sigma), \tag{11}$$

where K is the Gaussian curvature and χ is the Euler characteristic. For S^2, $\chi = 2$, so $\int K\, dA = 4\pi$.

The curvature cost has two factors:

(i) The *curvature load*: the ratio of token curvature quanta to the topological capacity of the bounding surface. The topological capacity of S^2 is $2\chi(S^2) = 4$ (the number of Gauss-Bonnet quanta in $4\pi = 2 \times 2\pi$). So the curvature load is $|\kappa|/(2\chi) = 4/4 = 1$.

(ii) The *geometric scale*: the bounding area measured in units of the Gauss-Bonnet quantum 2π. This gives $A/(2\pi) = 4\pi\lambda^2/(2\pi) = 2\lambda^2$.

The curvature cost is their product:

$$\boxed{J_{\text{curv}}(\lambda) = \frac{|\kappa|}{2\chi(S^2)} \cdot \frac{A}{2\pi} = 1 \cdot 2\lambda^2 = 2\lambda^2.} \tag{12}$$

Remark 4.2. Every factor in this formula is forced: $|\kappa| = 4$ by the cube structure (Proposition 4.1), $\chi(S^2) = 2$ by topology, $A = 4\pi\lambda^2$ by spherical geometry, and 2π by Gauss-Bonnet. No normalization is chosen by hand.

J_{curv} is quadratic in λ, not in a ratio of lengths—the mechanism that breaks scale invariance and allows the balance against the fixed bit cost to select an absolute length.

5 Main Derivation: The Cost-Balance Extremum

5.1 The Two Competing Costs

A single interaction event embedded in a causal diamond of spatial edge λ incurs two costs:

1. **Bit cost J_{bit}.** The information cost of the interaction event itself. In normalized cost accounting,
$$J_{\text{bit}} = 1. \tag{13}$$
The unnormalized form is $J_{\text{bit}} = \ln\varphi \approx 0.481$ (the natural information content of a φ-bit, as defined in (9)). The normalization absorbs $\ln\varphi$ into the cost unit.

2. **Curvature cost $J_{\text{curv}}(\lambda)$.** The geometric cost from Section 4: $J_{\text{curv}}(\lambda) = 2\lambda^2$.

5.2 The Total Cost Functional

$$C(\lambda) = J_{\text{bit}} + J_{\text{curv}}(\lambda) = 1 + 2\lambda^2. \tag{14}$$

5.3 The Minimal-Overhead Balance Condition

The minimal-overhead principle requires the discrete lattice to operate at the scale where neither cost channel carries unnecessary surplus. The total cost $C(\lambda)$ sums a scale-independent term (bit cost) and a scale-dependent term (curvature cost, growing as λ^2). The equilibrium is at the crossover:

$$J_{\text{bit}} = J_{\text{curv}}(\lambda_0). \tag{15}$$

If $\lambda > \lambda_0$, the curvature cost exceeds the bit cost and the lattice wastes geometric capacity. If $\lambda < \lambda_0$, the bit cost exceeds the curvature cost and the lattice wastes information capacity. The minimal-overhead principle excludes both regimes.

Theorem 5.1 (Recognition length in dimensionless units)**.** *The unique positive solution to* $J_{\text{bit}} = J_{\text{curv}}(\lambda_0)$ *is* $\lambda_0 = 1/\sqrt{2}$.

Proof. $J_{\text{curv}}(\lambda) = 2\lambda^2$ is strictly monotone increasing for $\lambda > 0$, and $J_{\text{bit}} = 1 > 0$. The equation $2\lambda^2 = 1$ has exactly one positive solution: $\lambda_0 = 1/\sqrt{2}$. □

Remark 5.2 (No circular dependence). This derivation uses only the bit cost (from φ-scaling) and the curvature cost (from Q_3 geometry via Gauss-Bonnet). It does not reference Newton's constant G, the Planck length ℓ_P, or any external physical input. G will be *defined* as a consequence in Section 7.

6 Dimensional Restoration and the Planck Gate Identity

6.1 From Dimensionless Balance to SI

The balance condition (15) is a statement in natural units, where $\lambda_{\text{rec}} = 1$, $c = 1$, and $\hbar = \varphi^{-5}$. The balance point $\lambda_0 = 1/\sqrt{2}$ is dimensionless. To convert to SI, we must determine what G—and hence the Planck length $\ell_P = \sqrt{\hbar G/c^3}$—equals.

The curvature-cost derivation *determines* G. The factor of π relating λ_{rec} to ℓ_P comes from Gauss-Bonnet.

6.2 The Gauss-Bonnet Bridge

The Gauss-Bonnet integral of S^2 is $\int K\, dA = 4\pi$, carried by $|\kappa| = 4$ curvature modes (Proposition 4.1). Each mode contributes a curvature load

$$\frac{\int K\, dA}{|\kappa|} = \frac{4\pi}{4} = \pi. \tag{16}$$

The Planck area $\ell_P^2 = \hbar G/c^3$ sets the area of one curvature quantum. Each mode carries π such quanta (in natural units where $\lambda_{\text{rec}} = 1$); the area per quantum is the inverse of the curvature load:

$$\ell_P^2 = \frac{1}{\pi} \quad \text{(natural units)}. \tag{17}$$

Equivalently: the voxel face area ($\lambda_{\text{rec}}^2 = 1$) contains π Planck cells per curvature mode.

6.3 Newton's Constant as a Derived Quantity

Definition 6.1 (Newton's constant from λ_{rec})**.** From $\ell_P^2 = \hbar G/c^3$ and (17):

$$\boxed{G = \frac{c^3\, \lambda_{\text{rec}}^2}{\pi\, \hbar}.} \tag{18}$$

In natural units ($\lambda_{\text{rec}} = 1$, $c = 1$, $\hbar = \varphi^{-5}$), this gives $G = \varphi^5/\pi$.

Corollary 6.2 (Einstein coupling constant)**.** *The Einstein gravitational coupling* $\kappa = 8\pi G/c^4$ *evaluates to*

$$\boxed{\kappa = 8\varphi^5.} \tag{19}$$

This is machine-verified as `kappa_einstein_eq` *in* `Constants.lean` *and* `kappa_rs_closed_form` *in* `Gravity/ZeroParameterGravity.lean` *[10].*

Proof. $\kappa = 8\pi G/c^4 = 8\pi \cdot \varphi^5/(\pi \cdot 1) = 8\varphi^5$ (using $G = \varphi^5/\pi$ and $c = 1$ in natural units). □

6.4 The Recognition Length in SI

Since $\ell_P^2 = \lambda_{\text{rec}}^2/\pi$ (from (17)):

$$\boxed{\lambda_{\text{rec}} = \sqrt{\pi}\,\ell_P = \sqrt{\frac{\pi\,\hbar G}{c^3}} \approx 2.86 \times 10^{-35}\ \text{m}.} \tag{20}$$

6.5 The Planck Gate Identity

$$\boxed{\frac{c^3\,\lambda_{\text{rec}}^2}{\hbar\,G} = \pi.} \tag{21}$$

Verification. Substituting $G = c^3\lambda_{\text{rec}}^2/(\pi\hbar)$: $c^3\lambda_{\text{rec}}^2/(\hbar \cdot c^3\lambda_{\text{rec}}^2/(\pi\hbar)) = \pi$. This identity is machine-verified in Lean 4 (Appendix A). □

7 Consequences

7.1 Position in the Forcing Chain

With λ_{rec} and G derived, the chain (1) continues to the fine-structure constant

$$\alpha^{-1} = 4\pi \cdot 11 - w_8 \ln\varphi - \frac{103}{102\pi^5} \approx 137.035 \quad \text{(Thomson limit)}, \tag{22}$$

and onward to the mass sector. Each step is algebraic once λ_{rec} is in place.

7.2 Quantum-Gravitational Bridge

λ_{rec} sits at the balance point of the curvature cost: the length scale where bit cost equals curvature cost. Below λ_{rec}, the $\hbar$ sector dominates. Above λ_{rec}, the G sector dominates. The reciprocal symmetry $J(x) = J(x^{-1})$ maps one regime to the other; λ_{rec} is the fixed point.

8 Independent Check: Bekenstein-Landauer Lower Bound

The Bekenstein-Hawking entropy bound $S \leq A/(4\ell_P^2)$ and the Landauer principle jointly provide an *independent* information-theoretic constraint on any fundamental length scale, derived without reference to the cost functional or Gauss-Bonnet geometry. An earlier version of this argument appears in [7].

Axiom 8.1 (One-bit causal diamond)**.** $S_\diamond = k_B \ln 2$.

Saturating the Bekenstein-Hawking bound gives the minimum bounding area capable of encoding one bit:

$$A_\diamond \;=\; 4\,\ell_P^2\,\ln 2. \tag{23}$$

The Bekenstein-Hawking formula is derived for spherical horizons, so the natural bounding geometry is the sphere S^2 of radius $\lambda_{\rm BL}$:

$$4\pi\,\lambda_{\rm BL}^2 \;=\; 4\,\ell_P^2\,\ln 2 \qquad\Longrightarrow\qquad \lambda_{\rm BL} \;=\; \sqrt{\frac{\ln 2}{\pi}}\,\ell_P \;\approx\; 7.6\times 10^{-36}\ \text{m}. \tag{24}$$

The curvature-extremum value is $\lambda_{\rm rec} = \sqrt{\pi}\,\ell_P \approx 2.86\times 10^{-35}$ m, a factor of

$$\frac{\lambda_{\rm rec}}{\lambda_{\rm BL}} \;=\; \frac{\pi}{\sqrt{\ln 2}} \;\approx\; 3.77 \tag{25}$$

above the Bekenstein-Landauer floor. This factor has a direct physical reading. The Bekenstein bound is an *inequality*: it sets the smallest sphere that *can* encode one bit. The cost-balance condition (Section **??**) selects the scale at which the lattice *does* operate. That the operating scale exceeds the information-theoretic minimum by a modest factor is expected and healthy: it means every recognition voxel carries comfortably more area than the Bekenstein minimum requires. The Bekenstein-Landauer route therefore serves as a consistency floor, not as a competing estimate of $\lambda_{\rm rec}$.

9 Phenomenological Consistency Checks

Via the effective recognition length $\lambda_{\rm eff} = \lambda_{\rm micro}\cdot f^{-1/4}$ [8]:

1. **Stellar luminosity balance:** $\lambda_{\rm eff}^{(\star)} = 6.3\times 10^{-5}$ m, $f^{(\star)} \simeq 3.2\times 10^{-122}$.
2. **Vacuum energy density:** $\lambda_{\rm eff}^{(\Lambda)} = 5.9\times 10^{-5}$ m, $f^{(\Lambda)} \simeq 3.4\times 10^{-122}$.

The two routes agree within 7%: $f = (3.3\pm 0.3)\times 10^{-122}$, $\lambda_{\rm eff} \approx 60\ \mu$m. These are validations, not inputs.

10 Discussion

10.1 Summary of the Derivation

Three ingredients, all internal to the derivation:

1. Bit cost $J_{\rm bit} = 1$ (normalized), from φ-scaling (Theorem 2.7).
2. Curvature cost $J_{\rm curv} = 2\lambda^2$, from the $|\kappa| = 4$ curvature quanta on Q_3 and the Gauss-Bonnet normalization on S^2 (Proposition 4.1, Section 4).
3. Balance condition $J_{\rm bit} = J_{\rm curv}$ from the minimal-overhead principle.

The balance selects $\lambda_0 = 1/\sqrt{2}$ in natural units. The Gauss-Bonnet bridge (Section 6.2) identifies $\ell_P^2 = 1/\pi$ in natural units, yielding $\lambda_{\rm rec} = \sqrt{\pi}\,\ell_P$ and $G = c^3\lambda_{\rm rec}^2/(\pi\hbar)$, with the Einstein coupling $\kappa = 8\varphi^5$.

10.2 Why Dimensional Closure Was Hard

$J(x)$ is scale-invariant: it depends only on ratios. The curvature-extremum argument breaks this invariance by introducing a cost quadratic in the physical edge length λ, not in a ratio. When balanced against the fixed bit cost, it selects a unique scale.

10.3 Scope and Assumptions

The derivation depends on three classes of input, listed here with their epistemic status:

(a) **Theorems.** The uniqueness of $J(x)$ (Theorem 2.3) and the forcing of φ, c, $\hbar$ along the derivation chain (1) are mathematical results, machine-verified in Lean 4 [10].

(b) **Framework results.** The curvature token count $|\kappa| = 4$ (Proposition 4.1) and the Gauss-Bonnet normalization on S^2 follow from the $D = 3$ cube topology Q_3, but rely on a specific mode-decomposition and equal-weight argument that some readers may classify as modeling assumptions rather than unavoidable consequences.

(c) **Postulates.** Two inputs are physical postulates, not theorems:

- The minimal-overhead balance condition $J_{\text{bit}} = J_{\text{curv}}$. This is the assertion that the lattice operates at the scale where informational and geometric costs are equal.
- The Planck-area bridge $\ell_P^2 = 1/\pi$ (Section 6.2), identifying the Planck area with the inverse of the Gauss-Bonnet curvature load per mode. This step connects the internal combinatorial geometry to the gravitational scale.

Given items (a)–(c), the output (λ_{rec}, G, κ) is unique with zero continuously adjustable parameters. The paper does not claim that these postulates are themselves derived from pure logic; it claims that, once accepted, they leave no parameter free to tune.

10.4 Curvature Scale versus Bekenstein Floor

The curvature-extremum route gives the operating scale $\lambda_{\text{rec}} = \sqrt{\pi}\,\ell_P \approx 2.86 \times 10^{-35}$ m. The Bekenstein-Landauer route (Section 8) gives an information-theoretic lower bound $\lambda_{\text{BL}} = \sqrt{\ln 2/\pi}\,\ell_P \approx 7.6 \times 10^{-36}$ m. The ratio $\lambda_{\text{rec}}/\lambda_{\text{BL}} = \pi/\sqrt{\ln 2} \approx 3.77$ is not a discrepancy but a consistency check: the Bekenstein bound is an inequality, and the lattice operating above this floor by a moderate algebraic factor is the expected regime. Both routes use spherical geometry (S^2) for the bounding surface, so the comparison involves no ad hoc geometric choice.

The curvature-extremum value is the canonical result because it follows from the cost functional and Gauss-Bonnet normalization alone, without external bounds (Bekenstein-Hawking, Landauer), and it is consistent with the machine-verified identity $\kappa = 8\varphi^5$ in the Lean formalization [10].

11 Discussion and Conclusion

With λ_{rec} derived, the framework determines both dimensionless and dimensional constants. The number of independent dimensional inputs drops from three (G, $\hbar$, c) to two ($\hbar$, c), with $G = c^3\lambda_{\text{rec}}^2/(\pi\hbar)$ and $\kappa = 8\varphi^5$. The factor of π originates in the Gauss-Bonnet curvature load per mode on S^2.

λ_{rec} is where the informational cost of an interaction event matches the geometric cost of the curvature it creates. Below it, $\hbar$ dominates; above it, G dominates. The reciprocal symmetry $J(x) = J(x^{-1})$ maps one regime to the other, and λ_{rec} is the fixed point. In the broader RS program, this scale is the prerequisite for addressing the mass spectrum: particle masses become expressible once a dimensional anchor is available.

All prerequisite theorems and the balance-condition derivation are machine-verified in Lean 4 with zero `sorry` statements (Appendix A).

A Lean 4 Machine Verification

The full Lean 4 codebase is at `https://github.com/jonwashburn/recognition-science` [10]. It contains 122 Lean files across 10 directories, with 32 verification certificates bundling related theorems. One axiom is used: `aczel_representation_axiom` in `Cost/FunctionalEquation.lean`, encoding the Aczél smoothness condition for the d'Alembert equation (a standard result; Aczél, 1966). All other results are proved unconditionally.

A.1 What Lean Verifies

The formalization covers:

(i) **Prerequisite theorems.** The uniqueness of J (Theorem 2.3), the forcing of φ (Theorem 2.7), and the forcing of $D = 3$ (Theorem 2.11), formalized with zero `sorry` statements.

(ii) **Balance-condition derivation.** The cost balance $J_{\text{bit}} = J_{\text{curv}}(\lambda)$, i.e., $1 = 2\lambda^2$, has the unique positive root $\lambda_0 = 1/\sqrt{2}$. This is formalized as `balance_at_lambda_0` and `balance_unique_positive_root` in `Constants/LambdaRecDerivation.lean` [10]. These proofs reference only the bit cost ($= 1$) and the curvature cost ($= 2\lambda^2$), *without* reference to G, ℓ_P, or any external constant.

(iii) **Algebraic consistency.** `Constants.lean` defines $G := \lambda_{\text{rec}}^2 c^3/(\pi\hbar)$ and verifies $\kappa = 8\varphi^5$ via `kappa_einstein_eq`. The Planck gate identity $c^3\lambda_{\text{rec}}^2/(\hbar G) = \pi$ is an algebraic consequence of this definition; its role is to confirm internal consistency.

A.2 What Lean Does Not Verify

The physical postulates listed in Section 10.3—the balance condition $J_{\text{bit}} = J_{\text{curv}}$ and the Planck-area bridge $\ell_P^2 = 1/\pi$—are *inputs* to the formalization, not outputs. Lean verifies the mathematical consequences of these postulates (e.g., that $\lambda_0 = 1/\sqrt{2}$ is the unique root, that

$\kappa = 8\varphi^5$ follows algebraically), but it does not establish that the postulates are physically inevitable. The question of whether the balance condition and the Planck-area identification are unavoidable consequences of discrete lattice dynamics, or whether they are substantive modeling choices, remains open.

A.3 Module Table

Table 1 lists the modules from the public repository [10] that formalize theorems cited in this paper.

Module path	**Key theorems**	**sorry**
`Cost/JcostCore.lean`	`Jcost_symm`, `Jcost_nonneg`, `Jcost_pos_of_ne_one`, `Jcost_strict_mono_on_one_infty`	0
`Cost/FunctionalEquation.lean`	`washburn_zlatanovic_uniqueness_aczel`, `dAlembert_to_ODE_theorem`, `ode_cosh_uniqueness_contdiff`	0
`Foundation/PhiForcing.lean`	`phi_forced`, `golden_constraint_unique`	0
`Foundation/EightTick.lean`	`phase_eighth_power_is_one`, `shift_period_8`	0
`Foundation/Dimension-Forcing.lean`	`eight_tick_forces_D3`, `dimension_forced`	0
`Constants.lean`	`phi_sq_eq`, `hbar_eq_phi_inv_fifth`, `G_pos`, `kappa_einstein_eq`	0
`Constants/LambdaRec-Derivation.lean`	`balance_at_lambda_0`, `balance_unique_positive_root`, `lambda_rec_is_forced`	0
`Gravity/ZeroParameter-Gravity.lean`	`kappa_rs_closed_form`, `kappa_bounds`, `equivalence_principle_automatic`	0

Table 1: Lean 4 modules from `github.com/jonwashburn/recognition-science` that formalize theorems cited in this paper. All paths are relative to `RecognitionScience/`.

The repository also contains 32 verification certificates (`Verification/` directory) that bundle related theorems. Of particular relevance: `NonCircularityCert` (verifies no experimental data enters upstream), `ZeroAdjustableParamsCert` (`knobsCount = 0`), and `ForcingChainCert` (the complete derivation chain in one structure).

A.4 Correspondence Between Paper and Lean

- **Theorem 2.3** (uniqueness of J): formalized as `washburn_zlatanovic_uniqueness_aczel` in `Cost/FunctionalEquation.lean`. The proof uses the Aczél axiom (`aczel_dAlembert_smooth`) to establish smoothness, then derives the ODE $H'' = H$ via `dAlembert_to_ODE_theorem`, and concludes $H = \cosh$ via `ode_cosh_uniqueness_contdiff`. The ODE uniqueness proof uses the integrating-factor diagonalization described in Section 2.1: `deriv_neg_self_zero` ($g' = -g \Rightarrow g = 0$) and `deriv_pos_self_zero` ($h' = h \Rightarrow h = 0$).

- **Theorem 2.7** (phi forcing): formalized as `phi_forced` and `golden_constraint_unique` in `Foundation/PhiForcing.lean`. The proof uses $\varphi^2 = \varphi + 1$ (from `phi_equation` in `Constants.lean`) and the algebraic identity $(r - \varphi)^2 \geq 0$ to force $r = \varphi$.

- **Theorem 2.11** ($D = 3$ forcing): formalized as `eight_tick_forces_D3` and `dimension_forced` in `Foundation/DimensionForcing.lean`. The proof is by case analysis on D.

- **Theorem 5.1** (balance condition): formalized as `balance_at_lambda_0` and `balance_unique_positi` in `Constants/LambdaRecDerivation.lean` [10]. The module defines:

$$\begin{aligned} J_{\text{bit}} &:= 1, \\ J_{\text{curv}}(\lambda) &:= 2\lambda^2, \\ \text{balanceResidual}(\lambda) &:= J_{\text{curv}}(\lambda) - J_{\text{bit}} = 2\lambda^2 - 1. \end{aligned}$$

  The proof establishes $\text{balanceResidual}(1/\sqrt{2}) = 0$ and uniqueness: for $\lambda > 0$, $\text{balanceResidual}(\lambda) = 0 \Leftrightarrow \lambda = 1/\sqrt{2}$. This is a *non-circular* verification: it uses only the cost functionals, without reference to G or ℓ_P.

- **Natural-unit constants** (Definition 2.12): all defined in `Constants.lean` [10]. Key definitions:

$$\begin{aligned} \texttt{phi} &:= (1 + \sqrt{5})/2, \\ \texttt{hbar} &:= \texttt{cLagLock} \cdot \texttt{tau0} = \varphi^{-5} \cdot 1, \\ \texttt{lambda_rec} &:= \texttt{ell0} = 1, \\ \texttt{G} &:= \texttt{lambda_rec}^2 \cdot \texttt{c}^3/(\pi \cdot \texttt{hbar}). \end{aligned}$$

  From these definitions, $\kappa = 8\pi G/c^4 = 8\varphi^5$ is proved as `kappa_einstein_eq`, and the Planck gate identity $c^3\lambda_{\text{rec}}^2/(\hbar G) = \pi$ follows algebraically.

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